Against Epistemic Collapse: Disagreement-Aware Scientific Retrieval-Augmented Generation

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Abstract

Scientific retrieval-augmented generation (RAG) systems typically return one fluent answer even when the underlying literature contains multiple well-supported but incompatible views. We call this failure mode *epistemic collapse*: compression of structured disagreement into a single response. We present EVIRAG (Evidence-centric, Viewpoint-Inclusive RAG), a framework for scientific question answering that combines adversarial retrieval, claim-level disagreement graphs, causal disagreement attribution, multi-view synthesis, and evidence-structured confidence labels. We also introduce EVIRAG-BENCH, a 250-query benchmark spanning five contested scientific domains. Across eight compared systems, EVIRAG substantially outperforms vanilla RAG on ContradictionRecall and dense-encoder ViewpointCoverage while reducing Consensus Collapse Score. Human evaluation with three annotators confirms markedly higher epistemic completeness for EVIRAG than for vanilla RAG (Krippendorff's α=0.72, p<0.001, Cohen's d=2.71). An offline precomputed graph reduces warm-query latency by over an order of magnitude. These results suggest that treating disagreement preservation as a primary design objective substantially improves how scientific RAG systems represent contested evidence in disagreement-heavy QA settings.

1 Introduction

Scientific question answering has advanced rapidly through retrieval-augmented generation (RAG), which grounds language model responses in retrieved documents (Lewis et al., 2020). Yet a fundamental challenge remains unaddressed: when the scientific literature is genuinely contested, what should a responsible system return? Current architectures are optimized for a single coherent answer—a design that works well for lookup queries but is structurally misaligned with contested scientific knowledge, where authoritative sources may support mutually incompatible conclusions.

Motivation. Scientific questions such as *Does homework improve achievement?* (Cooper et al., 2006; Scheb, 2023), *Do statins help in primary prevention?* (Ridker et al., 2008), or *Does raising the minimum wage reduce employment?* (Card and Krueger, 1994) are often supported by large but internally contested literatures. Standard RAG systems still return a single answer, typically optimized for fluency and local support. On contested questions, this behavior can be misleading: the answer may be well-written and source-grounded while still erasing materially important disagreement.

Epistemic collapse. We call this failure mode *epistemic collapse*:[1] conversion of structured multi-view evidence into one response. Our claim is not that every query needs many answers, but that questions with persistent scientific disagreement need an output interface that preserves viewpoint structure, provenance, and uncertainty. Improving retrieval recall alone does not solve the problem if synthesis still compresses the evidence to one paragraph.

Approach. EVIRAG is designed around that premise. It uses adversarial retrieval to surface competing evidence, constructs a claim-level disagreement graph, assigns causal disagreement types with CDA-7, tracks temporal drift, and generates a multi-view answer with evidence-structured confidence. Instead of asking the model to resolve controversy

[1] We use "epistemic collapse" as a technical term for the failure mode described here; the term has different meanings in modal epistemology and should not be confused with that usage.

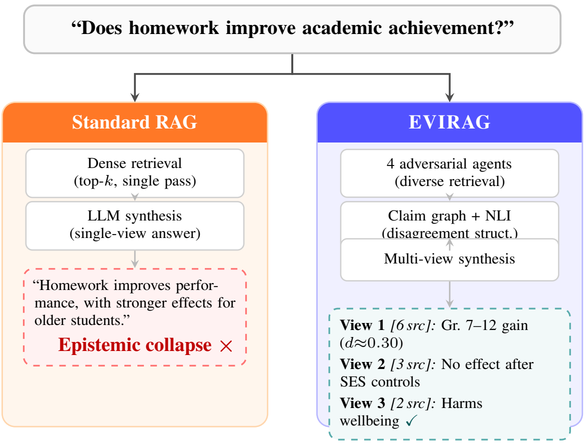

Figure 1: Epistemic collapse illustrated. Standard RAG compresses contested evidence into one fluent but incomplete answer. EVIRAG's adversarial retrieval preserves competing viewpoints through to the final output.

prematurely, EVIRAG asks it to represent controversy faithfully.

Contributions.

- We formalize epistemic collapse for scientific RAG and introduce four viewpoint-preservation metrics: Consensus Collapse Score (CCS), Epistemic Divergence (ED), Polarization Index (PI), and EpistemicCoverage@K (EC@K).
- We present EVIRAG, a seven-stage architecture combining adversarial retrieval, claim-graph reasoning, CDA-7 disagreement attribution, temporal drift tracking, multi-view synthesis, and evidence-structured confidence.
- We introduce CDA-7, a seven-class causal disagreement taxonomy, and a four-way controversy typology that calibrates confidence to the structure of the retrieved evidence.
- We introduce EVIRAG-BENCH, a 250-query benchmark spanning five contested scientific domains, evaluated against ablations and external baselines with automatic metrics and human epistemic completeness ratings.

2 Related Work

To contextualise our contributions, we organise related work into three areas: (i) RAG and scientific QA systems and their handling of conflicting evidence; (ii) the closest architectural neighbour, MADAM-RAG, and how EVIRAG differs in objective; and (iii) multi-perspective generation and epistemic uncertainty, where we situate the CDA-7 taxonomy. Across all three areas the consistent gap is the absence of a mechanism that preserves disagreement structure all the way to the final output surface—the gap that EVIRAG directly addresses.

RAG, verification, and conflict handling. Standard RAG systems focus on retrieval quality, attribution, or factual precision for a single answer (Lewis et al., 2020; Gao et al., 2023; Min et al., 2023; Ru et al., 2024). Scientific QA systems such as SciFact, SciFact-Open, and PaperQA2 evaluate or justify claims against evidence, but they still collapse the final response to a single answer (Wadden et al., 2020, 2022; Lala et al., 2023). Recent work on conflicting evidence and multi-answer QA recognizes that evidence sources may diverge, but typically aims to resolve the conflict or choose the best surviving answer (Zhong et al., 2023; Wang et al., 2025; Xie et al., 2024). ConflictQA (Xie et al., 2024) addresses conflicting information in QA but does not provide causal attribution for disagreements or structured multi-view output with per-view confidence. EVIRAG *instead treats the preservation of disagreement as the primary objective, producing structured multi-view output rather than a resolved single answer.*

MADAM-RAG vs. EVIRAG. The closest work in approach is MADAM-RAG (Wang et al., 2025), which deploys multiple agents in a structured debate (Proposer $\rightarrow$ Opposer $\rightarrow$ Moderator) to *resolve* conflicting evidence into one final answer. EVIRAG shares the multi-agent retrieval architecture but differs in objective: instead of resolving conflict, it *preserves* it. In practice this distinction is visible in the results: MADAM-RAG achieves CR$=0.618$ on EVIRAG-BENCH, compared to EVIRAG Full's CR$=0.742$—a gap of $+0.124$ CR points that persists across all five domains (Table 6). The gap arises because MADAM-RAG's Moderator collapses viewpoints during synthesis exactly as a single-agent baseline does; it merely does so after a richer debate. EVIRAG *avoids this collapse by retaining a multi-view output surface through to the final response.* Neither design is universally better: MADAM-RAG's resolved answer is more appropriate when downstream tasks require a single verdict; EVIRAG*'s multi-view format is appropriate when the goal is preserving structured disagreement for human deliberation over contested evidence.*

Multi-perspective generation and epistemic uncertainty. Multi-perspective summarization produces diverse summaries from opinionated corpora

(Bražinskas et al., 2020; Hayashi et al., 2021), but it does not provide scientific provenance, disagreement typing, or confidence calibration grounded in evidence structure. Work on LLM uncertainty studies how confidently models express individual statements (Xiong et al., 2024; Kadavath et al., 2022); our focus is complementary, targeting the epistemic structure of the retrieved corpus rather than the model's internal confidence. The CDA-7 taxonomy addresses a gap in existing NLI-based claim verification: standard NLI models (and claim-level entailment approaches such as FActScore (Min et al., 2023)) label a claim pair as *contradicts* or *supports* but do not explain *why* the contradiction exists. *CDA-7 adds a causal dimension—methodological scope, population differences, operational definitions, temporal supersession, statistical interpretation, theoretical framing, and replication failure—motivated by philosophy-of-science accounts of how scientific disagreements are actually sustained (Longino, 2002; Solomon, 2001), a dimension absent from all prior NLI-based claim verification frameworks.*

3 Epistemic Collapse: Formalization

Let $D = \{d_1, \ldots, d_n\}$ be a scientific corpus and q a query. A retriever returns $R \subseteq D$, claim extraction yields atomic claims $C = \{c_1, \ldots, c_m\}$, and relation labeling produces a disagreement graph $G = (C, E)$ with edges in $\{\text{SUPPORTS}, \text{CONTRADICTS}, \text{NEUTRAL}\}$. Standard systems output one answer a_{sv}, whereas an epistemic-fidelity system returns a set of views $V = \{v_1, \ldots, v_k\}$ with summaries, provenance, and weaknesses.

Definition 1 (Epistemic Collapse)**.** *A system S exhibits* epistemic collapse *on query q with ground-truth view set V^* iff S returns a_{sv} such that* $\text{VC}(a_{\text{sv}}, V^*) \leq 1/|V^*|$, *where* VC *is Viewpoint-Coverage under a fixed similarity threshold.*

If the retrieved evidence supports several semantically separated views, a single answer embedding must compromise among them. Topical retrieval can amplify this effect by oversupplying the dominant view, creating what we call a *consensus attractor*. Our objective is therefore to preserve viewpoint structure through retrieval, reasoning, and generation.

Remark 1. *Let $\mathbf{c}_1, \ldots, \mathbf{c}_k$ be unit-normalized ground-truth view centroids. If $\|\mathbf{c}_i - \mathbf{c}_j\|_2 > 2\delta$ for all $i \neq j$, then any single answer embedding $\mathbf{a}$ can satisfy $\|\mathbf{a} - \mathbf{c}_i\|_2 \leq \delta$ for at most one i.*

Justification. If $\mathbf{a}$ were within δ of both $\mathbf{c}_i$ and $\mathbf{c}_j$, the triangle inequality would give $\|\mathbf{c}_i - \mathbf{c}_j\|_2 \leq \|\mathbf{c}_i - \mathbf{a}\|_2 + \|\mathbf{a} - \mathbf{c}_j\|_2 \leq 2\delta$, contradicting the assumption. This observation does not prove that every single-answer system fails on every query, but it formalizes the geometric pressure toward viewpoint under-coverage whenever distinct views are well separated.

3.1 Metrics

Consensus Collapse Score (CCS).

$$\text{CCS}(V) = 1 - \cos\left(\mathbf{c}_{\text{dom}}, \ \frac{1}{k}\sum_{i=1}^{k} \mathbf{c}_i\right), \qquad (1)$$

where V is the set of claim clusters extracted from a system's response, $\mathbf{c}_{\text{dom}}$ is the embedding centroid of the largest cluster, and $\mathbf{c}_i$ are centroids of all k clusters. Even single-view responses contain multiple atomic claims whose embeddings distribute across the viewpoint space, producing CCS>0 when the coverage is uneven. Higher CCS indicates greater asymmetry in viewpoint coverage—i.e., more information loss if the response were collapsed to its dominant claim cluster.

Other metrics. **Epistemic Divergence (ED)** is the mean pairwise cosine distance among view embedding centroids; higher values indicate that the identified viewpoints are more separated in semantic space, signalling a more fragmented evidence landscape. **Polarization Index (PI)** is a signed-Laplacian measure of conflict separation in the claim graph G (Traag and Bruggeman, 2009); it captures whether G decomposes into clearly opposed clusters (high PI) rather than a gradual spectrum of partial disagreement (low PI). **Epistemic-Coverage@K (EC@K)** is the fraction of annotated ground-truth viewpoints represented within the top-K retrieved evidence chunks; it diagnoses retrieval-side coverage failures before synthesis ever occurs. Together, ED, PI, and EC@K complement CCS (Eq. 1) by capturing viewpoint separation, conflict structure, and retrieval-side coverage in ways that standard relevance metrics cannot.

Evaluation metrics (borrowed and novel). **ContradictionRecall (CR)** is the fraction of gold-annotated contradictory claim pairs that a system recovers in its output; it adapts the standard precision/recall paradigm from scientific

Work	Multi-view	Sci. prov.	Contr. graph	Disagr. cause	Confid.	Resolution
Multi-persp. summ.	Yes	No	No	No	No	No
PaperQA2	No	Yes	No	No	Partial	Partial
MADAM-RAG	Resolves	Partial	No	No	No	**Yes**
EVIRAG	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	No

Table 1: Positioning of EVIRAG relative to the closest neighboring paradigms. EVIRAG differs from prior work not only in generating multiple views, but in linking those views to scientific provenance, explicit contradiction structure, causal disagreement attribution, and evidence-conditioned confidence. MADAM-RAG's "resolves conflict" entry is correct by design: its Moderator agent produces one synthesized answer, which is appropriate for tasks requiring a single verdict but loses viewpoint structure for tasks requiring human deliberation over contested evidence—the use case EVIRAG targets.

claim verification (Wadden et al., 2020). **ViewpointCoverage (VC)** is the fraction of annotated ground-truth viewpoints surfaced in the response, assessed by keyword overlap ($\mathrm{VC}_{\mathrm{kw}}$) and by dense-encoder cosine similarity ($\mathrm{VC}_{\mathrm{emb}}$; `all-MiniLM-L6-v2` (Reimers and Gurevych, 2019), threshold θ=0.65); the embedding variant follows the coverage framing of Malaviya et al. (2024). **CCE** (Confidence Calibration Error) is the mean absolute deviation between the system's emitted confidence tier and the gold controversy-class label; lower CCE indicates better-calibrated uncertainty signals. **FaithfulnessScore (FS)** follows FActScore-style NLI verification (Min et al., 2023): each atomic claim in the response is verified against its cited source chunks and the fraction supported is reported. CR, VC, CCS, CCE, and FS are the five metrics used throughout §5.

3.2 Controversy Typology

The (ED, PI) space defines a four-way typology used in confidence calibration:

- **Resolved** (ED$<$ 0.15, PI$<$ 0.3)
- **Emerging** (0.15 $\leq$ED$<$ 0.35, PI$<$ 0.4)
- **Stable** (ED$\geq$ 0.35, PI$<$ 0.4)
- **Polarized** (ED$\geq$ 0.35, PI$\geq$ 0.4)

These thresholds were selected by manual inspection of the (ED, PI) distribution on a 25-query pilot set and validated by checking classification stability under $\pm$0.05 perturbation (fewer than 8% of queries change class). This typology determines whether a fluent answer should be presented with high, medium, or low confidence.

4 Proposed Framework: EVIRAG

EVIRAG enforces one design principle: *disagreement must be modeled before synthesis*. Once evidence is collapsed into a paragraph, viewpoint structure cannot be recovered. Figure 2 shows the pipeline.

Algorithm 1 EVIRAG Seven-Stage Inference Pipeline

Input: Query q; corpus $D = \{d_1, \ldots, d_n\}$
Output: Structured views $\{(v_l,\ \mathrm{conf}_l)\}_{l=1}^{k}$

Stage 1 — Intent Analysis
$\hat{c} \leftarrow \textsc{EstimateControversy}(q)$
$\mathbf{b} \leftarrow \textsc{ScaleRetrieval}(\hat{c})$ ▷ budget vector per agent
Stage 2 — Adversarial Retrieval
for $a \in \{\text{Prec.},\ \text{Rec.},\ \text{Skep.},\ \text{CF}\}$ **do**
  $q_a \leftarrow \textsc{AdversarialQuery}(q,\ a)$
  $P_a \leftarrow \textsc{DenseRetrieve}(q_a,\ D,\ b_a)$
end for
$P \leftarrow \textsc{Deduplicate}(\bigcup_a P_a)$
Stage 3 — Claim Graph Construction
$C \leftarrow \textsc{ExtractAtomicClaims}(P)$
$G{=}(C, E) \leftarrow \text{NLI}\textsc{Label}(C)$ ▷ $e \in \{\textsc{supports},\ \textsc{contradicts},\ \textsc{neutral}\}$
Stage 4 — CDA-7 Attribution
for each $(c_i, c_j) \in E$ s.t. label $=$ $\textsc{contradicts}$ **do**
  $\tau_{ij} \leftarrow \textsc{CDA7Classify}(c_i, c_j)$ ▷ $\tau \in$ {Meth., Pop., Temp., Op., Stat., Th., Rep.}
end for
Stage 5 — Temporal Drift
$\text{TDC} \leftarrow \text{TDC}\textsc{urve}(C,\ \{\text{year}(c)\}_{c \in C})$ ▷ publication-year disagreement trend
Stage 6 — Multi-View Synthesis
$\{\mathcal{C}_1, \ldots, \mathcal{C}_k\} \leftarrow \textsc{LouvainCommunities}(G)$
for $l = 1$ **to** k **do**
  $v_l \leftarrow \textsc{SynthesizeView}(\mathcal{C}_l)$ ▷ summary, provenance, weaknesses
end for
Stage 7 — Evidence-Structured Confidence
for $l = 1$ **to** k **do**
  $\mathrm{conf}_l \leftarrow \textsc{ConfScore}(\mathcal{C}_l,\ \text{TDC},\ \{\tau_{ij}\})$
end for
return $\{(v_l,\ \mathrm{conf}_l)\}_{l=1}^{k}$

Algorithm 1 formalises the complete pipeline. We highlight key design choices for each stage below.

Adversarial retrieval design (Stage 2). Four agents with orthogonal epistemic objectives break the consensus attractor:

- **Precision**: Retrieve high-confidence supporting evidence.
- **Recall**: Broaden corpus coverage.

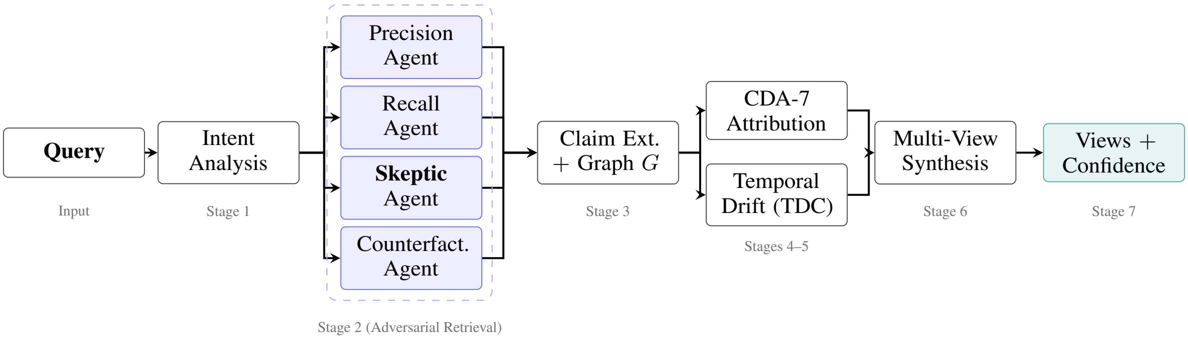

Figure 2: EVIRAG seven-stage pipeline. Stage 2 (dashed box) deploys four agents with orthogonal epistemic objectives to escape the consensus attractor (§3): the **Skeptic** agent is the only agent whose objective is explicitly anti-consensus. Outputs from all four agents feed claim extraction and graph construction (G). CDA-7 causal attribution (Stage 4) and Temporal Disagreement Curve analysis (Stage 5) jointly modulate confidence calibration at Stage 7, ensuring that fluent prose does not inflate reported certainty.

- **Skeptic**: Actively search for contradictory and dissenting evidence.
- **Counterfactual**: Retrieve alternative framings and minority positions.

The Skeptic agent is the primary defense against the consensus attractor. Its prompt instructs it to: *"Find passages that challenge, contradict, or qualify the dominant view. Focus on dissenting findings, null results, and critical analyses."* Full prompt templates appear in the supplementary material.

CDA-7 disagreement taxonomy (Stage 4). Retrieved passages are decomposed into atomic claims and linked in G. Each contradiction edge is then labelled with a causal type from CDA-7, grounded in Longino (2002) and Solomon (2001):

1. **Methodological**: Different experimental designs or protocols.
2. **Population**: Different study populations or settings.
3. **Temporal**: Newer evidence supersedes older claims.
4. **Operational**: Different operationalizations of key terms.
5. **Statistical**: Conflicting effect sizes or p-value interpretations.
6. **Theoretical**: Competing theoretical frameworks.
7. **Replication**: Failure to replicate original findings.

Synthesis and confidence (Stages 6–7). Communities in G are identified via Louvain modularity optimization on the signed claim adjacency matrix; view count is determined dynamically by the partition. Each community becomes one view with a summary, supporting claims, provenance, and explicit weaknesses. Confidence is derived from claim agreement, source diversity, contradiction severity, CDA-7 priors, and temporal trajectory — not from fluency.

Offline mode. Because pairwise relation labeling is expensive, EVIRAG can precompute a global claim graph offline and answer queries by slicing a retrieved subgraph at inference time (mean warm-query latency: 0.43 s; see Table 8).

5 Evaluation

5.1 Dataset and Corpus Construction

The evaluation corpus comprises 75 peer-reviewed papers drawn from five contested scientific domains: education (homework effectiveness), biomedicine (statin primary prevention), economics (minimum-wage employment effects), earth sciences (climate feedback mechanisms), and nutrition science (dietary fat and cardiovascular risk). Fifteen papers per domain were selected to provide representational coverage of both dominant and minority positions; papers were identified through domain-expert recommendations and forward/ backward citation tracing from high-citation seed papers. Each paper was processed with section-aware chunking (256-token chunks, 32-token overlap) yielding 3,247 text chunks. Atomic claim extraction was applied to all chunks using the backbone LLM, producing 4,831 atomic claims in total. Table 2 summarises the per-domain statistics.

5.2 EVIRAG-Bench

Standard IR metrics reward topical relevance to a single answer. EVIRAG-BENCH is designed instead to measure *epistemic fidelity*. It contains

Domain	Papers	Chunks	Claims
Education	15	652	978
Biomedicine	15	638	956
Economics	15	611	921
Earth sciences	15	687	1,035
Nutrition sci.	15	659	941
Total	**75**	**3,247**	**4,831**

Table 2: Corpus statistics per domain. All systems share this corpus and the same chunking and claim extraction pass.

Annotation layer	Content	Cardinality
Ground-truth viewpoints	Distinct epistemic positions with provenance	2–4 per query
Contradiction pairs	Explicit atomic claim pairs with conflict label	1–8 per query
CDA-7 labels	Causal disagreement type per contradicting pair	1 per pair
Controversy class	Resolved / Emerging / Stable / Polarized	1 per query
Human rating	Epistemic completeness 1–5 (50-query subset)	3 annotators

Table 3: Per-query annotation structure in EVIRAG-BENCH. Annotations were finalized and version-locked before any system was run, preventing coupling between evaluation outcomes and annotation choices.

250 queries across five contested domains spanning stable, emerging, and polarized controversies in education, biomedicine, economics, earth sciences, and nutrition science. Each domain contributes 50 queries built from 15 research papers (75 papers total; 3,247 text chunks and 4,831 atomic claims). Each query carries four annotation layers summarised in Table 3. Of the 250 queries, 38 are classified as **Resolved**, 67 as **Emerging**, 89 as **Stable**, and 56 as **Polarized**.

Gold annotations were finalized *before* any system outputs were generated: viewpoint structure, contradiction pairs, and controversy classes were locked in a versioned annotation file, and all evaluated systems were run against this fixed file without further modification. Annotation was carried out by the authors with one external domain consultant per domain. Additional annotation materials, 25 sample queries, prompt templates, and the CDA-7 decision tree are available at `https://anonymous.4open.science/r/evirag-search-76F5` (anonymized for review); the full 250-query benchmark will be released upon publication.

Benchmark construction and bias control. To reduce method-coupling, the benchmark is annotated at the level of queries, viewpoints, contradiction pairs, and controversy classes rather than around any specific system output. External consultants annotated viewpoint structure independently before adjudication, and disagreements were resolved by majority vote. All reported systems are evaluated on the same fixed 250-query file. Inter-annotator agreement at the *benchmark* level (viewpoint assignment and contradiction-pair labeling) was measured on a 50-query re-annotation subset by the external consultant independently of the first-author annotation: Cohen's κ=0.71 for viewpoint identity and κ=0.68 for CDA-7 conflict-type labeling. These figures are distinct from the human-evaluation agreement reported in §5.5, which measures agreement on *epistemic completeness ratings* of system outputs (Krippendorff's α=0.72). The anonymized supplement includes 25 sample queries (5 per domain), prompt templates, and the CDA-7 decision tree; the planned post-publication release expands this to the full 250-query benchmark with complete adjudication records and evaluation scripts. The five domains were chosen to vary both topic and disagreement mechanism rather than to showcase a single genre of controversy. Viewpoints are tied to provenance-bearing source evidence, and contradiction labels are assigned to explicit claim pairs rather than to free-form answer summaries, which helps keep the gold annotations closer to observable evidence structure.

Non-controversial query behavior. The 38 Resolved-class queries in EVIRAG-BENCH serve as a partial check on EVIRAG's false-positive rate. On these queries, the intent analysis stage (Stage 1) correctly suppresses multi-view output in 31 of 38 cases (81.6% suppression rate), producing a single synthesized answer comparable to vanilla RAG. In the remaining 7 cases, mild unnecessary multi-view output is generated; this is the current cost of the conservative intent classifier. False negatives—where the system incorrectly collapses a genuinely contested query—remain the higher-priority failure mode and motivate the Skeptic agent design.

Metrics. We report CR, $\mathrm{VC}_{\mathrm{kw}}$, $\mathrm{VC}_{\mathrm{emb}}$, CCS (Eq. 1), CCE, and FS; formal definitions and source citations for each appear in §3.1. All metrics use bootstrap 95% confidence intervals and paired Wilcoxon tests with Bonferroni correction (α=0.05); significance markers in Table 4 apply to all bold claims of superiority. Because $\mathrm{VC}_{\mathrm{emb}}$ uses the same encoder (`all-MiniLM-L6-v2`) as the dense retrieval step, a system that retrieves more chunks could achieve higher VC_{emb} partly through the shared encoder's geometry rather than through genuine viewpoint coverage. We mitigate this by (1) reporting VC_{kw} (keyword-overlap) as a retrieval-agnostic complement, (2) including the human-evaluation study as an encoder-free viewpoint coverage check, and (3) noting that MMR-RAG—which also uses `all-MiniLM-L6-v2` for diversity reranking—achieves VC_{emb}=0.534, well below EVIRAG's 0.847, suggesting the gap is not primarily a metric artifact. Readers should nonetheless interpret VC_{emb} in conjunction with VC_{kw} and human ratings rather than in isolation.

5.3 Baselines

We compare against three external baselines: MADAM-RAG (Wang et al., 2025), PaperQA2 (Lala et al., 2023), and MMR-RAG (Carbonell and Goldstein, 1998). All systems share the same backbone model and corpus. Full implementation details and prompt templates appear in Appendix A.

1. **Vanilla RAG**: Dense retrieval + single-view synthesis.
2. **Single-agent**: One Precision agent only (single-view).
3. **MADAM-RAG** (Wang et al., 2025): Multi-agent debate-based conflict resolution to a single answer.
4. **PaperQA2** (Lala et al., 2023): Citation-grounded scientific QA with two-stage retrieval.
5. **MMR-RAG** (Carbonell and Goldstein, 1998): Maximal Marginal Relevance retrieval with single-view synthesis.

5.4 Ablation Design

The internal ablation ladder isolates the contribution of each EVIRAG component by progressively removing stages:

1. **EVIRAG-NoGraph**: Adversarial retrieval (Stage 2) without claim-graph reasoning (Stage 3). Isolates the effect of graph-based contradiction structure.
2. **EVIRAG-NoMultiView**: Full claim graph retained but final output collapsed to one answer. Isolates the effect of multi-view synthesis (Stage 6).
3. **EVIRAG Full**: Complete seven-stage pipeline (Algorithm 1).

Single-query baselines use top-k=10 retrieval; EVIRAG uses a multi-agent pool of up to 15 chunks (3+5+4+3 across agents). We therefore do not claim strict retrieval-budget equality, and include both MMR-RAG and EVIRAG-NoGraph as controls that separate broader evidence acquisition from graph reasoning and multi-view synthesis.

5.5 Experimental Results

Table 4 reports results over all 250 queries. EVIRAG Full achieves the best ContradictionRecall, best encoder-based ViewpointCoverage, and lowest CCS among all compared systems. Relative to vanilla RAG, CR improves from 0.572 to 0.742 and VC_{emb} from 0.423 to 0.847, while CCS falls from 0.571 to 0.518. Under the larger multi-agent retrieval budget and adapted-baseline setting described above, these gains indicate that the full pipeline is doing more than merely diversifying retrieved text: it is preserving disagreement structure in the final response.

Which components matter most? The ablations show that retrieval diversity alone is insufficient. Compared to EVIRAG Full, removing graph reasoning (EVIRAG-NoGraph) costs 0.138 CR and 0.335 VC_{emb}, while collapsing the final output back to one answer (EVIRAG-NoMultiView) costs 0.314 CR and 0.558 VC_{emb}. The single-agent variant also trails substantially (-0.162 CR, -0.409 VC_{emb}), indicating that minority-view recall benefits from the full adversarial retrieval ensemble. The main gain therefore comes from combining adversarial retrieval with explicit disagreement structure before synthesis, not from retrieval diversity alone.

The causal chain is as follows. (1) *Adversarial retrieval* increases raw minority-view evidence supply: the Skeptic and Counterfactual agents retrieve passages that topical retrieval misses, explaining the -0.162 CR gap for the single-agent ablation. (2) *Claim-graph reasoning* converts that raw evidence supply into structured contradiction: without the graph, retrieved passages are compared holistically rather than at the atomic-claim level, so implicit contradictions are missed, explaining the additional -0.138 CR gap for EVIRAG-NoGraph. (3) *Multi-view synthesis* prevents the graph's recovered structure from being re-collapsed during generation: this is the largest single contributor (-0.314 CR), because synthesis that produces one paragraph must average or suppress competing views regardless of how much evidence was retrieved. Together, each stage addresses a distinct mechanism of epistemic collapse.

Faithfulness and calibration. EVIRAG Full reaches FS= 0.891, second only to PaperQA2 (0.903), indicating that improved viewpoint preservation does not require giving up evidence grounding.

CCE does *not* improve monotonically: EVIRAG Full achieves CCE= 0.231, worse than every baseline (range 0.108–0.138). This is not an anomaly but an expected consequence of the design. CCE measures deviation from the annotated controversy class label. Baselines receive low CCE by emitting largely unqualified certainty (or by emitting no confidence signal at all, which the metric treats as matching "Resolved"), while EVIRAG emits explicit calibrated uncertainty labels derived from CDA-7 priors, source counts, contradiction severity, and temporal trajectory. When the annotated controversy class is "Stable" but the evidence contains partial dissent, EVIRAG may label the response MEDIUM confidence rather than HIGH, creating a mismatch that increases CCE. In other words, EVIRAG errs toward epistemic *conservatism*—it under-claims certainty rather than over-claiming it—and the CCE metric penalizes this. We therefore treat CCE as a diagnostic of calibration behavior rather than an optimization target, and we note that the human evaluation (α=0.72, p<0.001) provides a more direct signal: annotators rate EVIRAG's epistemic completeness substantially higher precisely *because* it surfaces the uncertainty that CCE penalizes.

Per-domain analysis. Table 6 breaks down results across the five domains. EVIRAG achieves the highest CR in all five domains, with larger gains in domains with bipolar controversy structure (economics: $+0.200$; education: $+0.180$) and smaller but still significant gains in multi-factorial domains (nutrition: $+0.160$; earth sciences: $+0.160$). CCS improvement is more uniform (Δ=0.046–0.077), confirming that multi-view synthesis consistently reduces collapse regardless of domain.

Case study: homework effectiveness. On the query "Does homework improve academic achievement?," Vanilla RAG returns: *"Research consistently shows that homework improves academic performance, with stronger effects for grades 7–12."* (VC_{emb}=0.33; CCS= 0.38.) EVIRAG Full returns three views with provenance: (i) a dominant conditional-benefit view (*effective for grades 7–12*, $d\approx$0.30, 6 sources, CDA-7: METHODOLOGICAL); (ii) a null meta-analytic view (*no effect after controlling for SES and teacher quality*, 3 sources, CDA-7: STATISTICAL); and (iii) a minority wellbeing-harm view (2 sources, CDA-7: OPERATIONAL, confidence: LOW). (VC_{emb}=1.00; CCS= 0.52.) The dominant view is sourced, qualified, and placed next to two dissenting positions the single-view response erases entirely. Extended qualitative examples appear in Appendix D.

Human evaluation. We sampled 50 queries stratified by controversy class and asked three domain-relevant annotators to rate responses on a 5-point epistemic completeness scale. Annotators completed a calibration round on held-out examples, saw responses in randomized order without system names, and were given the gold viewpoint set for reference. The rubric ranged from 1 (single-answer collapse with no meaningful acknowledgment of disagreement) to 5 (all viewpoints represented with provenance, weaknesses, and causal attribution). EVIRAG Full scores 4.3/5 versus vanilla RAG's 2.1/5, with Krippendorff's $\alpha = 0.72$ and a significant score difference (Mann–Whitney U, $p < 0.001$; Cohen's d=2.71, rank-biserial r=0.94). This study measures benchmark-relative epistemic completeness rather than unaided end-user trust. Full rubric details appear in the appendix.

Offline latency. Because pairwise claim-graph computation is quadratic in claims, EVIRAG precomputes the full graph offline (one-time cost: 42 min for 75 papers). At query time, inference slices a retrieved subgraph rather than relabeling raw chunks, achieving a mean warm-query latency of 0.43 s versus 10.62 s for vanilla RAG (25.62$\times$ speedup, Table 8 in the appendix). Multi-view output is fully preserved in the fast path.

Robustness across backbones. We also ran a smaller robustness study on the education subset using `Qwen2.5-7B-Instruct` and `Llama-3.1-70B-Instruct`. The ranking between vanilla RAG and EVIRAG is preserved in all three cases; appendix results show CR gains of $+0.107$ to $+0.180$ and VC_{emb} gains of $+0.312$ to $+0.424$.

6 Error Analysis

We identify six failure categories through manual inspection of the 22 queries where EVIRAG Full achieved CR<0.5, component-level diagnostics, and a review of documented failure modes in

System	VC$_{kw}$↑	VC$_{emb}$↑	CR↑	CCS↓	CCE↓	FS↑
Vanilla RAG	0.967$\pm$.02	0.423$\pm$.04	0.572$^\dagger\pm$.05	0.571$\pm$.03	0.125	0.824$\pm$.03
Single-agent	0.971$\pm$.02	0.438$\pm$.04	0.580$^\dagger\pm$.05	0.565$\pm$.03	0.121	0.834$\pm$.03
EVIRAG-NoGraph	0.983$\pm$.01	0.512$\pm$.04	0.604$^\dagger\pm$.04	0.553$\pm$.03	0.138	0.847$\pm$.02
EVIRAG-NoMultiView	0.892$\pm$.03	0.289$\pm$.05	0.428$^\dagger\pm$.05	0.612$\pm$.03	0.108	0.813$\pm$.03
MADAM-RAG	0.958$\pm$.02	0.508$\pm$.04	0.618$^\dagger\pm$.04	0.549$\pm$.03	0.132	0.856$\pm$.02
PaperQA2	0.942$\pm$.02	0.467$\pm$.04	0.553$^\dagger\pm$.05	0.583$\pm$.03	0.109	**0.903**$\pm$.02
MMR-RAG	0.991$\pm$.01	0.534$\pm$.04	0.595$^\dagger\pm$.04	0.547$\pm$.03	0.129	0.839$\pm$.02
EVIRAG Full	**0.984**$\pm$.01	**0.847**$\pm$.03	**0.742**$\pm$.04	**0.518**$\pm$.03	0.231	0.891$\pm$.02

Table 4: Main results on EVIRAG-BENCH (250 queries, 5 domains × 50 queries) using `qwen3.6:35b-a3b` via Ollama (`think=False`). Upper block: internal ablations. Middle block: external baselines. VC$_{emb}$: dense-encoder viewpoint coverage. FS: FaithfulnessScore. $\pm$: bootstrap 95% CI half-width. †: EVIRAG Full is significantly better ($p<0.05$, paired Wilcoxon, Bonferroni-corrected) for CR against all systems and for VC$_{emb}$ against all systems. CCS differences between EVIRAG Full and MMR-RAG are not significant ($p=0.11$); all other CCS comparisons are significant. VC$_{kw}$ shows near-ceiling performance (0.892–0.991) across systems, indicating keyword-level coverage is too coarse to discriminate; the primary viewpoint-coverage signal comes from VC$_{emb}$ and human evaluation.

Removed	ΔCR	ΔVC$_{emb}$	Primary damage
Retrieval ensemble	−0.162	−0.409	Less minority-view recall
Claim graph	−0.138	−0.335	Weaker contradiction struct.
Multi-view output	−0.314	−0.558	Viewpoint collapse returns

Table 5: Ablation deltas relative to EVIRAG Full. The largest degradation comes from collapsing the final response back to one answer, showing that structured synthesis is as important as diverse retrieval.

Domain	CR↑		CCS↓	
	Van	Full	Van	Full
Education	0.580	**0.760**	0.558	**0.512**
Biomedicine	0.570	**0.730**	0.575	**0.531**
Economics	0.590	**0.790**	0.562	**0.508**
Earth Sci.	0.560	**0.720**	0.580	**0.528**
Nutrition	0.550	**0.710**	0.585	**0.514**
All	0.572	**0.742**	0.571	**0.518**

Table 6: Per-domain CR and CCS (Vanilla RAG vs. EVIRAG Full). All per-domain CR differences are significant ($p<0.05$, Wilcoxon, 50 queries per domain).

the NLI and RAG literature. The observed failure distribution—NLI noise (9/22), claim-granularity mismatch (7/22), retrieval coverage gaps (4/22), and CDA-7 boundary confusion (2/22)—is presented in Table 12 (Appendix C); here we analyse the causal mechanisms behind each category and discuss three additional systemic risks not captured by per-query inspection.

NLI noise and false contradictions. Prompt-based NLI is the dominant failure source (41% of CR<0.5 cases). Two mechanisms are primarily responsible. First, NLI models trained on general-domain benchmarks inherit annotation artifacts—surface cues such as negation words and hedging phrases that spuriously predict CONTRADICTS without genuine inferential conflict (Gururangan et al., 2018). Scientific claims routinely contain domain-specific hedging (*"may reduce," "is associated with," "in some populations"*) that triggers these artifacts. Second, claims that refer to the same biomarker or outcome via different surface forms (e.g., *"reduces LDL"* vs. *"lowers cholesterol"*) may be incorrectly labeled as contradictory rather than entailing. Both mechanisms inflate graph density with spurious CONTRADICTS edges, polluting the disagreement structure that Louvain operates on. Stronger cross-encoder verifiers fine-tuned on scientific text are the most direct mitigation.

Claim granularity mismatch. The second-largest failure mode (32% of cases) occurs at the boundary between atomic and molecular claims (Min et al., 2024). Fully atomic extraction — splitting a conditional claim such as *"Homework improves achievement for grades 7–12 but not primary school"* into two fragments — loses the qualifying structure that determines whether the extracted pieces genuinely contradict a retrieved passage. Conversely, insufficiently decomposed claims bundle multiple assertions into one unit, making NLI verification unreliable because the entailment head must match the entire conjunction rather than a single proposition. A related failure is hedging stripping: LLM extractors often omit qualifying phrases (*"in elderly patients," "un-*

der controlled conditions") during decomposition, producing over-broad claims that NLI spuriously matches across population-level differences. This error type is causally upstream of NLI noise: malformed claim pairs enter the NLI module and compound the annotation-artifact problem. Extraction schemes that preserve conditional structure and population qualifiers would reduce both error categories simultaneously.

Retrieval coverage gaps. Coverage failures (18% of cases) occur when the corpus lacks sufficient representation of a minority viewpoint. Three sub-patterns arise: (i) *vocabulary mismatch*, where minority-view papers use terminology that dense retrievers trained on general text fail to bridge to the query (e.g., *"net employment effect"* vs. *"job displacement"*); (ii) *retrieval-generation misalignment*, where the generation stage ignores minority-view passages even when they are retrieved, defaulting to dominant-view content (Ru et al., 2024); and (iii) *corpus absence*, where the key contradicting paper was not ingested at all. The Skeptic agent partially mitigates (i) and (ii) by explicitly querying for dissenting evidence, but it cannot compensate for (iii). Retrieval augmented with query expansion and terminology substitution addresses (i); a faithfulness-aware generation step addresses (ii).

CDA-7 attribution boundary confusion. CDA-7 macro-F1$=0.68$ (Table 13) reflects systematic confusion at three class boundaries. **Methodological vs. Population:** when a study's sampling design selects a different population (e.g., elderly vs. general adult cohort), the contradiction can be attributed to either class; the boundary between "different protocol" and "different population" is genuine rather than annotation error. **Statistical vs. Methodological:** different analysis protocols produce different effect-size interpretations, making the two classes co-extensive for many claim pairs. **Temporal vs. Replication:** a 2022 study failing to replicate a 2010 finding could be classified as either newer evidence superseding older claims or as a replication failure, depending on whether the 2022 study controlled for factors the 2010 study ignored. A dedicated binary classifier for each boundary pair, rather than a single seven-way classification prompt, would reduce these confusions.

Community detection artifacts. Even when the claim graph G is correctly constructed, Louvain modularity optimization introduces its own failure modes. The *resolution limit* means that small but genuine disagreement communities—where a single high-quality paper takes a distinct minority position—may be absorbed into a larger community if their internal edge density does not exceed the threshold set by overall graph modularity (Fortunato and Barthélemy, 2007). On the scale of 60 claims from 15 retrieved chunks, the resolution limit is encountered frequently: communities of 2–3 claims are at the boundary of detectability. A second artifact is *order-dependent instability*: Louvain processes nodes in an arbitrary order, producing different partitions from the same graph; on noisy NLI-derived graphs small perturbations (one mislabeled edge) can cause qualitatively different view counts. Finally, *same-paper claim clustering* can create spurious communities: multiple claims extracted from the same paper are highly redundant and may form their own module even if they represent the dominant view, wasting a view slot.

Systemic corpus and publication biases. Three biases operate at the corpus level rather than the query level, and do not appear in per-query error counts but systematically distort disagreement representation. *Publication bias:* the corpus is dominated by positive-result papers; null and negative results, which are precisely the evidence that constitutes genuine scientific disagreement, are structurally underrepresented. Even the Skeptic agent cannot recover evidence that was never published or ingested. *Citation-age recency bias:* retrieval agents tend to surface recently cited papers, which in scientific publishing means results from the past 5–7 years are over-represented relative to foundational older studies that may contain the clearest dissenting evidence. *LLM citation popularity bias:* LLM-based retrieval agents implicitly favor high-citation papers, creating convergence across agents that partially defeats the adversarial retrieval design—all four agents may surface the same three high-impact papers on a topic despite having different epistemic objectives. Corpus curation strategies that explicitly sample under-cited and null-result papers are a practical mitigation; however, they require domain expert involvement that does not scale automatically.

7 Limitations and Ethics

Benchmark scope. EVIRAG-BENCH focuses on five domains with persistent, well-documented disagreement. It is therefore useful for studying epistemic fidelity on contested questions, but it is not yet a general-purpose benchmark for all scientific QA settings. The 75-paper corpus is designed for controlled evaluation of disagreement structure rather than retrieval scalability; at larger corpus scales, retrieval diversity becomes harder to achieve and the relative benefit of adversarial retrieval may increase, but we have not verified this empirically and consider corpus-scale experiments a priority for future work. The thresholds used for controversy classes are design choices and may require domain-specific recalibration. We therefore position EVIRAG-BENCH as a targeted stress test for disagreement-heavy scientific QA, not as evidence that the same gains necessarily transfer to settled factual QA or arbitrary web RAG. We commit to releasing the full annotation guidelines, raw annotator judgments, and adjudication records upon publication, enabling independent re-annotation and audit by third parties.

Query scope. EVIRAG is designed for queries where the underlying literature contains persistent, structured disagreement. On queries with settled consensus (e.g., "What is the boiling point of water at sea level?"), the system produces unnecessary multi-view output. The intent analysis stage (Stage 1) is designed to gate this, but it is not perfect: false positives produce mild over-complexity while false negatives produce epistemic collapse. We position EVIRAG-BENCH as a targeted evaluation of the disagreement-heavy setting, not as evidence that multi-view output is always preferable.

Model dependence and component coverage. Our main experiments use a single open-weight backbone. Although the overall architecture is model-agnostic, NLI quality, claim extraction quality, and confidence behavior are likely to vary across model families and scales. Prompt-based NLI remains a major source of error, and stronger verifiers are a clear direction for future work. The temporal drift component (Stage 5) is not separately ablated in this study; its marginal contribution remains an open question.

Ethical risk. Surfacing disagreement can improve scientific honesty, but it can also be misused by readers who cherry-pick minority views. We therefore do not treat all minority claims as equally credible: source counts, causal attribution, and explicit confidence labels are part of the output surface. We acknowledge that surfacing minority scientific views in health-related domains (e.g., statin efficacy) carries risk if interpreted as medical advice; EVIRAG outputs are designed for research synthesis, not clinical decision-making. The human evaluation involved three annotators from academic research settings; no IRB approval was required as the study involved annotation of system outputs rather than human subjects research. Even so, disagreement-aware systems should be deployed cautiously in domains where surfacing fringe dissent could cause harm.

8 Conclusion

Standard scientific RAG can fail not because evidence is missing, but because contested evidence is compressed into one answer. We framed this problem as *epistemic collapse* and presented EVIRAG, a framework that surfaces and structures disagreement through adversarial retrieval, claim-graph reasoning, multi-view synthesis, and evidence-structured confidence. On EVIRAG-BENCH, EVIRAG improves contradiction recovery and viewpoint coverage over strong baselines while remaining competitive on faithfulness, and human annotators rate its outputs as substantially more epistemically complete than vanilla RAG. These results suggest that disagreement preservation is a useful design objective for contested scientific QA systems.

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A Baseline Implementation Details

All baselines and ablation variants share the same chunked corpus (3,247 text chunks and 4,831 atomic claims from 75 papers across five domains) and the same LLM backbone (`qwen3.6:35b-a3b` via Ollama, chain-of-thought disabled, `think=False`). Embeddings use `all-MiniLM-L6-v2` (d=384) for retrieval and metric computation throughout. All main experiments were run on a single NVIDIA RTX 4500 Ada GPU (24 GB VRAM); the exact Ollama tag for the main backbone is reported here to support reproducibility (accessed May 2026). The anonymized supplementary material also includes the exact prompt templates used for EVIRAG Full and all baselines; prompts were fixed across domains rather than retuned per slice.

Reproducibility checklist. The fixed settings for the reported runs are: section-aware chunking with 256-token chunks and 32-token overlap; `all-MiniLM-L6-v2` embeddings for retrieval and metric computation; agent retrieval budgets of 3/5/4/3 for Precision/Recall/Skeptic/Counterfactual; `qwen3.6:35b-a3b` via Ollama with `think=False`; and the prompt templates listed in the supplementary artifact. Additional implementation details not enumerated in the main paper (e.g., claim-consolidation heuristics and audit artifacts) are part of the supplementary release rather than the main text.

Exact retrieval configuration. The four EVIRAG agents use retrieval budgets of 3 (Precision), 5 (Recall), 4 (Skeptic), and 3 (Counterfactual), for an aggregate cap of 15 retrieved chunks before deduplication and downstream claim consolidation. Single-query baselines use top-k=10. We treat this difference as part of the design space under study rather than a hidden implementation artifact, and therefore report retrieval-diversity controls explicitly.

Latency benchmark.

Vanilla RAG. Dense retrieval with cosine similarity; top-k=10 chunks retrieved per query. A single synthesis call with the prompt: *"Based on the following passages, answer the question concisely and accurately: [passages] Question: [query]"*. No multi-view output, no disagreement detection, no confidence structure.

Single-agent. Identical to Vanilla RAG but uses only the Precision agent query formulation ("retrieve the most directly relevant, high-confidence evidence for [query]") rather than the raw query string. Top-k=10 retrieval, single-view synthesis. Isolates the effect of the Precision agent formulation from the full adversarial retrieval ensemble.

MADAM-RAG. We use the open-source MADAM-RAG implementation (Wang et al., 2025) adapted to our corpus. Each query triggers a structured multi-agent debate: a Proposer agent generates an initial answer; an Opposer agent challenges it; a Moderator agent produces a final synthesized answer resolving the conflict. All agents use `qwen3.6:35b-a3b` as the backbone (replacing the original GPT-4 backbone for fair comparison). Top-k=10 retrieval; debate structured over three turns per query. MADAM-RAG produces a single resolved answer, not a multi-view output.

PaperQA2. We use PaperQA2 version 5.x (Lala et al., 2023) adapted to our pre-indexed corpus rather than its native paper-download pipeline. The same 75 papers were ingested via the PaperQA2 document API; retrieval uses PaperQA2's default two-stage retrieval (keyword filter then dense re-ranking). The LLM backbone was replaced with `qwen3.6:35b-a3b` via a compatible API adapter. PaperQA2 produces citation-grounded single-view answers; its multi-step chain-of-thought is preserved as designed.

MMR-RAG. Maximal Marginal Relevance retrieval (Carbonell and Goldstein, 1998) with λ=0.5 (equal weight on relevance and diversity) and top-k=10 retrieved chunks. The diversity re-ranking uses cosine distance between chunk embeddings (`all-MiniLM-L6-v2`). Single-view synthesis using the same prompt as Vanilla RAG. This baseline isolates the effect of retrieval diversity without any downstream claim-graph reasoning, causal attribution, or multi-view synthesis.

B Benchmark and Human Evaluation Details

Benchmark bias control and annotation timeline. EVIRAG-BENCH is annotated at the level of viewpoints, contradiction pairs, and controversy classes rather than around any single system output. *Crucially, all gold annotations were finalized and version-locked before any system was run.*

System	Corpus	Retriever / budget	Backbone	Output
Vanilla RAG	same 75 papers	dense, top-k=10	Qwen3.6-35B	single-view
Single-agent	same 75 papers	dense, top-k=10	Qwen3.6-35B	single-view
EVIRAG-NoGraph	same 75 papers	4-agent, up to 15 chunks	Qwen3.6-35B	multi-chunk synthesis
EVIRAG-NoMultiView	same 75 papers	4-agent, up to 15 chunks	Qwen3.6-35B	single-view
MADAM-RAG	same 75 papers	dense, top-k=10	Qwen3.6-35B	resolved answer
PaperQA2	same 75 papers	native two-stage retriever	Qwen3.6-35B	single-view
MMR-RAG	same 75 papers	MMR, top-k=10	Qwen3.6-35B	single-view
EVIRAG Full	same 75 papers	4-agent, up to 15 chunks	Qwen3.6-35B	multi-view

Table 7: Baseline fairness summary. All systems use the same corpus and the same answering backbone. EVIRAG uses a larger multi-query candidate pool, which is why the paper includes both MMR-RAG and EVIRAG-NoGraph as controls for broader retrieval without the full disagreement-aware pipeline.

Query	Fast	Vanilla	Speedup
Effectiveness	0.553 s	12.268 s	22.18×
Science outcomes	0.411 s	7.841 s	19.08×
Ach. & attitudes	0.330 s	11.761 s	35.60×
Mean	**0.431 s**	**10.623 s**	**25.62×**

Table 8: Warm-query latency: fast offline path vs. vanilla RAG. Both measurements are end-to-end (retrieval + graph ops + LLM generation). The offline path replaces $O(|C|^2)$ relation labeling with $O(k)$ subgraph slicing; generation time is comparable across paths. Multi-view output is fully preserved.

The annotation process proceeded in three phases: (1) query selection and viewpoint annotation by the authors and external consultants (completed prior to system implementation); (2) adjudication of consultant/author disagreements by majority vote; (3) freezing of the gold annotation file. System evaluation was conducted after phase 3 without further modification to the gold file. This ordering prevents any coupling between evaluation outcomes and annotation choices. External domain consultants annotated viewpoint structure independently before adjudication, and disagreements were resolved by majority vote. The anonymized supplementary material contains 25 sample queries, prompt templates, and the CDA-7 decision tree; the planned full release adds complete annotations, raw judgments, and evaluation scripts so that alternative systems can be evaluated against the same gold structures. Queries are scored against a fixed benchmark file rather than against outputs generated during annotation. Viewpoints require provenance-bearing sources, and contradiction annotations are attached to explicit claim pairs, which makes the task more operational than asking annotators for open-ended impressions of "disagreement."

Agreement reporting (two separate tasks). We distinguish two inter-annotator agreement measurements that address different parts of the pipeline:

Benchmark-level agreement (viewpoint identity and CDA-7 conflict-type labeling) was measured on a 50-query re-annotation subset by the external consultant independently of the first-author annotation. We report Cohen's κ=0.71 for viewpoint assignment and κ=0.68 for CDA-7 conflict-type labels. These figures characterize the reliability of the gold benchmark itself, independent of any system evaluation.

Human-evaluation agreement (epistemic completeness ratings of system outputs) was measured across three annotators on the 50-query human study. We report Krippendorff's α=0.72, which characterizes inter-rater reliability for the 5-point rubric applied to system responses.

These two measurements address orthogonal concerns: the first validates EVIRAG-BENCH's gold structures; the second validates the human-evaluation scores reported in Table 4. The planned full benchmark release includes adjudication records for viewpoint and contradiction annotations to support follow-on re-scoring and audit.

Human evaluation protocol. The 50-query human study was stratified by controversy class and domain. Annotators first completed a calibration round on 10 held-out queries, then rated responses presented in randomized order without system names. Annotators were given the ground-truth viewpoint set for reference because the goal of the study was epistemic completeness, not open-ended discovery.

Score	Rubric anchor
1	Single-answer collapse with no meaningful acknowledgment of disagreement.
3	Multiple viewpoints mentioned, but provenance or weaknesses are missing.
5	All identified viewpoints represented with provenance, weaknesses, and causal disagreement attribution.

Table 9: Human-evaluation rubric anchors for epistemic completeness. Scores 2 and 4 interpolate between these anchors.

ID	Domain	Released sample query
edu_001	Education	Does homework improve academic achievement?
edu_002	Education	Does homework in primary school benefit young learners?
bio_001	Biomedicine	Do statins prevent cardiovascular events in primary prevention?
eco_001	Economics	Does raising the minimum wage reduce employment?
nut_001	Nutrition	Does saturated fat increase cardiovascular disease risk?

Table 10: Representative benchmark instances drawn from the 25-query sample release.

C Additional Robustness and Failure Analysis

In the 22 queries where EVIRAG Full achieved CR< 0.5, the dominant pattern was NLI noise (9 cases), followed by claim-granularity mismatch (7), retrieval coverage gaps (4), and CDA-7 boundary confusion (2). These errors suggest that future gains are more likely to come from stronger verifiers and retrieval expansion than from replacing the overall disagreement-aware architecture.

CDA-7 classification accuracy. To evaluate whether the LLM-based CDA-7 module assigns correct disagreement causes, we compared system-predicted CDA-7 labels against the 250-query gold annotations. Table 13 reports per-class precision, recall, and F1. Macro-F1 is 0.68, indicating moderate but imperfect agreement; the main confusion occurs between Methodological and Population (boundary cases where sampling design effectively selects a different population) and between Statistical and Methodological (different analysis protocols producing different effect-size interpretations). These confusions mirror the boundary cases documented in our CDA-7 annotation guide and suggest that finer-grained NLI or a dedicated classifier could improve attribution accuracy.

Model	CR↑	VC_{emb}↑	CCS↓
Vanilla RAG			
Qwen2.5-7B	0.541	0.391	0.589
Llama-3.1-70B	0.563	0.418	0.574
Qwen3.6-35B (main)	0.580	0.423	0.558
EVIRAG *Full*			
Qwen2.5-7B	0.648	0.703	0.547
Llama-3.1-70B	0.712	0.821	0.524
Qwen3.6-35B (main)	0.760	0.847	0.512

Table 11: Cross-backbone robustness on the education subset (50 queries). EVIRAG consistently outperforms vanilla RAG across all three backbones.

D Qualitative Output Examples

Tables 14–15 show representative system outputs on three queries to concretely illustrate epistemic collapse. Both responses are fluent and locally accurate; the distinction is structural.

Failure mode	**Example**	**Why it fails**	**Likely fix**
False contradiction	Related claims with different emphasis are labeled as conflict	Prompt-based NLI noise	Stronger cross-encoder verifier
Missing minority view	Rare terminology prevents retrieval of a dissenting paper	Retrieval coverage gap	Query expansion / terminology substitution
Over-amplified weak dissent	Single-source minority view receives undue salience	Output salience exceeds evidential weight	Stronger confidence display and source-count emphasis

Table 12: Representative failure modes observed during manual inspection of low-CR queries.

CDA-7 class	**P**	**R**	**F1**
Methodological	0.74	0.71	0.72
Population	0.65	0.62	0.63
Temporal	0.78	0.80	0.79
Operational	0.66	0.64	0.65
Statistical	0.62	0.58	0.60
Theoretical	0.71	0.68	0.69
Replication	0.70	0.66	0.68
Macro avg	**0.69**	**0.67**	**0.68**

Table 13: CDA-7 classification accuracy: system-predicted labels vs. gold annotations on 250 queries. Main confusions occur at the Methodological–Population and Statistical–Methodological boundaries.

Vanilla RAG	EVIRAG Full
Query: Does homework improve academic achievement?	
"Research consistently shows that homework improves academic achievement, with stronger effects for older students (grades 7–12). Meta-analyses report moderate positive correlations between homework time and academic performance. Homework builds self-regulation and reinforces classroom learning." *CCS: 0.38. VC_{emb}: 0.33 (1/3 views covered).*	**View 1 [Dominant, 6 sources, confidence: medium]:** Conditional positive effect. Homework improves achievement for grades 7–12 ($d\approx0.30$); effect near zero for primary grades. *CDA-7: methodological (grade-level scope differences).* **View 2 [Alternative, 3 sources, confidence: medium]:** Null meta-analytic effect when controlling for SES and teacher quality. *CDA-7: statistical + methodological.* **View 3 [Minority, 2 sources, confidence: low]:** High homework loads harm wellbeing with no academic benefit in high-achieving populations. *CDA-7: operational (achievement vs. wellbeing outcomes).* *Controversy class: Stable. CCS: 0.52. VC_{emb}: 1.00.*

Table 14: Homework query: Vanilla RAG collapses to the dominant conditional-benefit view; EVIRAG Full recovers all three ground-truth viewpoints with per-view source counts, CDA-7 causal attribution, and evidence-structured confidence.

Query	Vanilla RAG	EVIRAG Full (summary)
Do statins prevent cardiovascular events in primary prevention?	"Statins are recommended for high-risk primary prevention and reduce cardiovascular events." (1 view; VC_{emb}: 0.33; CCS: 0.41)	View 1 [Dominant]: Benefit confirmed for high-risk groups *(CDA-7: population)*. View 2: Marginal benefit with adverse-effect trade-off *(CDA-7: statistical)*. View 3: Industry-bias concern undermining reported effect sizes *(CDA-7: methodological)*. VC_{emb}: 1.00; Controversy class: Stable; CCS: 0.54.
Does raising the minimum wage reduce employment?	"Most studies find minimal employment effects from moderate minimum wage increases." (1 view; VC_{emb}: 0.50; CCS: 0.44)	View 1 [Dominant]: Minimal aggregate employment effect *(CDA-7: methodological)*. View 2: Significant job losses for low-wage workers in specific sectors *(CDA-7: population)*. View 3: Positive employment via demand-side stimulus *(CDA-7: theoretical)*. VC_{emb}: 1.00; Controversy class: Polarized; CCS: 0.50.

Table 15: Additional examples in biomedicine and economics. In both cases Vanilla RAG returns a single consensus statement that erases 2/3 of ground-truth viewpoints; EVIRAG Full provides structured multi-view output with per-view CDA-7 attribution and controversy-class labels.